

1

$I = \iiint_E z \, dv$

$E = \{ (x, y, z) : 0 \leq x \leq 1, 3x \leq y \leq 3, 0 \leq z \leq \sqrt{9-y^2} \}$

$I = \int_0^1 \int_{3x}^3 \int_0^{\sqrt{9-y^2}} z \, dz \, dy \, dx$

$= \int_0^1 \int_{3x}^3 \frac{1}{2} (9-y^2) \, dy \, dx = \frac{1}{2} \int_0^1 (9(3-3x) - \frac{1}{3}(27-27x^3)) \, dx$

$= \frac{1}{2} \int_0^1 (18 - 27x + 9x^3) \, dx = \frac{1}{2} (18 - \frac{27}{2} + \frac{9}{4}) = \frac{27}{8}$

2.

$z = \sqrt{x^2+y^2}$
 $z = \sqrt{2-x^2-y^2}$

$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} xy \, dz \, dy \, dx$

$= \iiint_E xy \, dv$

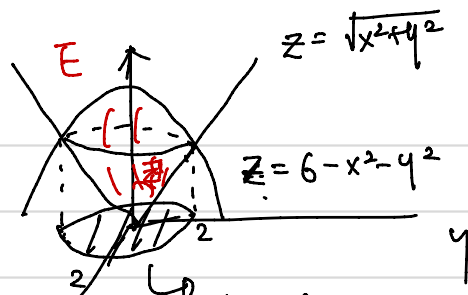
where $E = \{ (\rho, \theta, \phi) : 0 \leq \rho \leq \sqrt{2}, 0 \leq \theta \leq \frac{\pi}{2}, 0 \leq \phi \leq \frac{\pi}{4} \}$

$\therefore \iiint_E xy \, dV = \int_0^{\sqrt{2}} \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{4}} \rho^2 \sin^2 \phi \cos \theta \sin \theta \, \rho^2 \sin \phi \, d\phi \, d\theta \, d\rho$

$= \left(\int_0^{\sqrt{2}} \rho^4 \, d\rho \right) \left(\int_0^{\frac{\pi}{2}} \cos \theta \sin \theta \, d\theta \right) \left(\int_0^{\frac{\pi}{4}} \sin^3 \phi \, d\phi \right)$

$= \dots = \frac{4}{15} \sqrt{2} - \frac{1}{3}$

$$3. I = \iiint_E xy \, dV$$

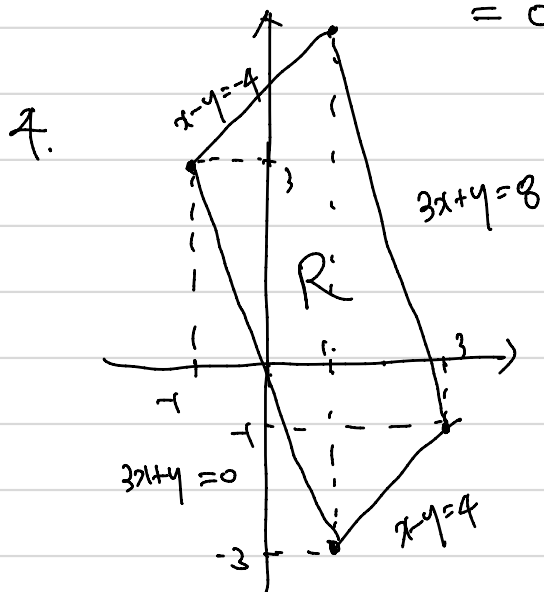


where

$$E = \{(r, \theta, z) : 0 \leq r \leq 2, 0 \leq \theta \leq 2\pi, r \leq z \leq 6 - r^2\}$$

$$I = \int_0^2 \int_0^{2\pi} \int_r^{6-r^2} r^2 \sin \theta \cos \theta \, r \, dz \, d\theta \, dr$$

$$= \left(\int_0^{2\pi} \sin \theta \cos \theta \, d\theta \right) \left(\int_0^2 \int_r^{6-r^2} r^3 \, dz \, dr \right) = 0.$$



$$\text{Set } \begin{cases} u = 3x + y \\ v = x - y \end{cases} \Rightarrow \begin{cases} x = \frac{1}{4}(u + v) \\ y = \frac{1}{4}(u - 3v) \end{cases}$$

Then R is the image of

$$\{(u, v) : 0 \leq u \leq 8, -4 \leq v \leq 4\}$$

w.r.t. the above mapping.

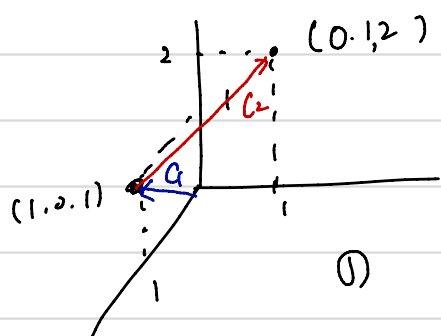
$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & -\frac{3}{4} \end{vmatrix} = -\frac{1}{4}$$

$\therefore$ By the change of variables

$$\iint_R (4x + 8y) \, dA = \int_0^8 \int_{-4}^4 ((u+v) + 2(u-3v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \, dv \, du$$

$$= \int_0^8 \int_{-4}^4 (3u - 5v) \frac{1}{4} \, dv \, du = \dots = 192.$$

5.



$$C_1 : \mathbf{r}(t) = \langle t, 0, t \rangle, \quad 0 \leq t \leq 1$$

$$C_2 : \mathbf{r}(t) = \langle 1, 0, 1 \rangle + t \langle -1, 1, 1 \rangle, \\ = \langle 1-t, t, 1+t \rangle \quad 0 \leq t \leq 1$$

$$\textcircled{1} \int_{C_1} (y+z) dx + (x+z) dy + (x+y) dz$$

$$= \int_0^1 ((0+t) \cdot 1 + (t+t) \cdot 0 + (t+0) \cdot 1) dt$$

$$= \int_0^1 2t dt = 1$$

$$\textcircled{2} \int_{C_2} (y+z) dx + (x+z) dy + (x+y) dz$$

$$= \int_0^1 ((t+1+t) \cdot (-1) + (1-t+1+t) \cdot 1 + (1-t+t) \cdot 1) dt$$

$$= \int_0^1 (-2t + 2) dt = 1$$

$$\therefore \underline{\underline{2}}$$